

Marcelo Soiber

Software Engineering | Full-Stack Development | Applied AI

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PROFESSIONAL PROFILE

Software engineering professional with over 9 years of experience in application development and payment technology. Background in Python and PHP, with recent projects in TypeScript, Node.js, Angular, and FastAPI. Deepening expertise in software engineering with applied AI and building practical solutions for semantic search, agents, and machine learning.

PROFESSIONAL EXPERIENCE

Software Engineering · gsurf

Santa Catarina, Brazil · Payment technology

Experience in software development for payment solutions, combining technical expertise with business problem-solving.

SELECTED PROJECTS

Knowledge Hub · AI knowledge base · Source code on GitHub

Application for organizing files and notes, performing semantic search, and answering questions from users' own documents. RAG architecture with an Angular frontend, FastAPI API, PostgreSQL/pgvector, local models through LM Studio or Ollama, and an MCP server for agent integration. Designed to preserve data control and privacy.

Credit Card Fraud Detection · Applied AI study

Laboratory that simulates the identification of anomalous behavior in financial transactions. Combines machine-learning models with generative AI and agents to support the interpretation of suspicious patterns and prioritize human review.

Knight's Tour · Genetic algorithms in a full-stack application

Implementation of a genetic algorithm to solve the knight's tour on a chessboard. Node.js and TypeScript backend with job execution and Server-Sent Events monitoring; interface with charts and SQLite persistence. Includes Warnsdorff's heuristic, mutation operators, and stagnation recovery.

Taxonomic identification system · Undergraduate capstone project, 2015

Pair-developed system to support the identification of taxonomic groups using dichotomous keys during the Computer Science degree at Unisul.

EDUCATION

Postgraduate Program in Software Engineering with Applied AI · UNIPDS · 2026–2027 (in progress)

Computer Science · University of Southern Santa Catarina (Unisul) · Capstone presented in 2015

TECHNICAL SKILLS

Development: Python, PHP, Java, JavaScript, TypeScript, Node.js, Angular, FastAPI, and APIs.

Architecture and tools: MCP, Server-Sent Events, Docker, Git, and full-stack application development.

Data and AI: PostgreSQL, pgvector, SQLite, embeddings, RAG, LLMs, machine learning, and genetic algorithms.

LANGUAGES AND RECOGNITION

Native Portuguese · Basic English

3rd place in Programming Marathon (2012 and 2013) · Participation in Unisul Tech Day (2011 and 2013)